Advanced log Analysis:

- Log correlation: In this if a single failed login appears it's a noise. five failed logins followed by a successful login on different machine followed by in a case of powershell execution that is a lateral movement attack.
- Anomaly detection: Instead of looking for bad things you look for not normal things. for example, a user downloads 50 MB a day, today they downloaded 5 GB.
- Log Enrichment: Raw logs are often mysterious. Enrichment adds details such as who, what, where, when etc.
  such as An IP address 10.0.2.15 is enriched with "Department: Finance" and Asset: SQL server".

Threat Intelligence Integration:

- Threat intelligence turns soc from reactive to proactive.
- The pyramid of pain: Advanced threat intelligence focuses on the top of the pyramics TTPs (tactics, techniques, procedures) rather than just easy indicators like hashes, or IP addresses which attackers change constantly.
- STIX/TAXII: Let's say in raw language these are the language and delivery methods.
  STIX: It's the format used to describe the threats (language).
  TAXII: It's the protocol used to move the information from a provider to your SIEM.
- Threat Hunting: You take a known TTP from the MITTRE ATT&CK framework and search your environment: such as where in my network is powershell being used in a way that matches this specific style of the threat giver.

Incident Escalation Workflows:

- A SOC is only as good as its communication. If a Tier 1 analyst finds a breach but doesn't know who to call or what information to provide, the attacker wins.
- **The Tiered Model:**
  Tier 1 (Triage): High volume, quick True/False positive determination.
  Tier 2 (Responder): Deep dive. They perform forensics and containment.

Tier 3 (Hunter/Forensics): Advanced malware analysis.

- **SITREP (Situation Report):** In the heat of a breach, leadership doesn't need technical jargon; they need a SITREP:

  Current Status: What is happening?

  Impact: What is stolen?

  Actions Taken: What have we done?

  Recommendations: What do we need next?

- **SOAR (Security Orchestration, Automation, and Response):** A SOAR playbook can automatically escalate a ticket to Tier 2 if it sees a high-severity alert from a user's laptop, saving precious minutes. This is a time saver technique and effort saver as well. It will escalate as soon as it detects alert without user intervention at first.